

Jasper Vermeulen

PhD Researcher | Human-Robot Collaboration | Socio-Technical Systems and Human Factors
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Keywords

Human-Robot Interaction; Human-Robot Collaboration; Ethnography; CSCW; Human Factors

Key Skills

- Human-Robot Collaboration (HRC)
- User Experience (UX) Research
- Human Factors & Ergonomics
- Qualitative Research & Analysis Methods
- Public Speaking
- Project Management & Planning
- Stakeholder Engagement & Industry Collaboration
- Academic Writing

Education

PhD in Human Factors of Human-Robot Collaboration

Queensland University of Technology, Australia | Expected Defence Date September 2026

Thesis Title: Understanding the Human(s) in Human-Robot Collaboration: A Socio-Technical Inquiry into Robot-Assisted Surgeries

Research Focus: Human factors in multi-actor Human-Robot Collaboration (HRC) within surgical settings, with an emphasis on the gap between how robot-assisted surgery has been designed and how it is performed in practice.

Key outcome: Based on three interrelated empirical studies, including video analysis of robot-assisted surgeries, 100+ hours of ethnographic observation in a Brisbane hospital, and interviews with surgical team members across Australian cities, this research reveals a significant gap between the designed and enacted realities of robot-assisted surgery. Using Socio-Technical Grounded Theory (STGT), these findings are translated into a novel, transferable theory of multi-actor HRC, applicable across healthcare, manufacturing, and construction.

Selected outputs: 4 journal publications, 6 conference publications, 6 workshop contributions, 8 invited lectures, and 1 podcast, drawn from 100+ hours of fieldwork across video analysis, observation, and interview studies.

Post-Master's in Human-Media Interaction

Keio University, Japan

Focus: Japanese Language + Emergent technologies in the context of mitigating loneliness.

MA in Human-Media Interaction (High Distinction)

Erasmus University, Netherlands

Thesis: User acceptance and attitudes towards smart home doorbells.

BSc in Media and Communication (High Distinction)

Erasmus University, Netherlands

Thesis: User attitudes towards smart home technologies. (Best Thesis Nominee)

Other Education

CHI2026: Doctoral Consortium Training

Remote Attendance | Issued April 2026

HRI2026: HRI Pioneers (Doctoral Consortium Training)

Remote Attendance | Issued March 2026

Professional Training for PhDs (2-day)

Australian Cobotics Centre, Australia | Issued October 2025

Effective Assessment and Feedback University Teaching

Queensland University of Technology, Australia | Issued April 2025

Approaches to Planning and Designing Learning University Teaching

Queensland University of Technology, Australia | Issued March 2025

OzCHI2024: Doctoral Consortium Training

University of Queensland, Australia | Issued November 2024

Human Factors in Healthcare: Designing Safe Systems Short Course

Monash University, Australia | Issued October 2024

Responsible Use of AI in Research Writing

Queensland University of Technology, Australia | Issued March 2024

ICH Good Clinical Practice E6 (R2)

The Global Health Network | Issued February 2024

Presenting Research Seminars

Queensland University of Technology, Australia | Issued September 2023

Research Project Management

Queensland University of Technology, Australia | Issued July 2023

Writing Style and Grammar for Researchers

Queensland University of Technology, Australia | Issued June 2023

Experience

PhD Researcher

Australian Cobotics Centre & Queensland University of Technology (QUT)

Brisbane, Australia | July 2023 – Present

- Conducting research on human factors and ergonomics in Human-Robot Collaboration (HRC) within surgical environments.
- Provided actionable insights to Stryker Inc. for Mako-specific product and workflow improvements (details classified).
- Presented research at national and international conferences, engaging both academic and industry audiences.

Research Assistant

Construction Architecture and Robotics Lab (CARL) & QUT School of Architecture and Built Environment

Brisbane, Australia | November 2025 – Present

- CARL is a new research initiative at QUT. Under the guidance of A/Prof Tim Rose and Prof Glenda Caldwell, I conducted a systematic scoping review (PRISMA) to investigate the adoption barriers and enablers of HRC in the construction industry.
- Conference Paper conditionally accepted for presentation at the 43rd International Symposium on Automation and Robotics in Construction (June 22-26, 2026) titled "Reconceptualising Human-Robot Collaboration in AEC: A PRISMA Scoping Review."
- Journal paper following the PRISMA review under development.

Sessional Academic (Tutor)

Queensland University of Technology, QUTYou010, People with Robots Unit

Brisbane, Australia | September 2024 – Present

- Delivering weekly, 2-hour tutorials and supporting student learning in human-robot interaction and design-related topics.
- Providing academic guidance and marking assessments.

- Involved in redesigning curriculum, leading to improved student engagement captured in student survey.

Research Assistant

Australian Cobotics Centre & QUT School of Architecture and Built Environment
Brisbane, Australia | November 2024 – November 2025

- Contributed to the “Brain-Computer Interface (BCI) for HRC” project in collaboration with Dr Alan Burden, exploring how BCI can support humans engaging in various HRC tasks.
- Project outcomes and deliverables classified.

Research Assistant

QUT Centre for Robotics & QUT Design Lab
Brisbane, Australia | January 2024 – November 2024

- Contributed to the “Robotic Blended Sonification” project in collaboration with A/Prof Stine Johansen, exploring how sound design can enhance human-robot collaboration.
- Assisted with study design, participant recruitment, and qualitative data collection and analysis.
- Co-presented project findings at the International Symposium on Electronic Art (ISEA) 2024, highlighting the intersection of robotics, sound, and user experience.
- Co-authored a publication on embodied composition methods for imagining robotic sound space, accepted at the ACM/IEEE HRI 2025 Conference.

Project Manager and Education Innovation Consultant

Erasmus University Rotterdam, Innovation Hub
August 2020 – August 2022

- Within the school of Culture, Communication and History, I assisted with tutorial facilitation, student assessment, and academic guidance.
- Advised course coordinators on how to make their classes more interactive, especially remotely conducted ones.
- Supervised a team of two student assistants to support faculty needs during COVID-19 and beyond.

Project Manager Intern

DPDK Digital Agency, Rotterdam
January 2020 – August 2020

- Oversaw several projects related to internal marketing (e.g. newsletter) and external marketing (e.g. SEO optimisation).
- I redesigned and implemented a new intranet platform for all employees.

Teaching Assistant Undergraduate and Graduate Courses

Erasmus University of Rotterdam

January 2020 – August 2022

- Created lecture videos for Media Technologies and Their Impacts.
- Taught Media Technologies and Their Impacts, Surveillance and Society, Qualitative Methods in the Social Sciences, Advanced Qualitative Methods in the Social Sciences and Emerging Media Technologies.

Research Software Skills

- Atlas.ti
- NVIVO
- Overleaf
- Microsoft Office: PowerPoint, Excel, Word, etc.
- Qualtrics Survey Software
- Mentimeter

Presentations

Reconceptualising Human-Robot Collaboration in AEC: A PRISMA Scoping Review

43rd International Symposium on Automation and Robotics in Construction (ISARC) 2026, National University Singapore, SG
June 2026

The Invisible Work of Robotic Surgery: How Specialists Support, Shoulder, and Sustain Human-Robot Collaboration

ACM DIS 2026, National University Singapore, SG
June 2026

The Choreography of Care: An Ethnographic Study of Human-Robot Collaboration in Makoplasty Surgeries

ACM CHI 2026, Remote Video Presentation
April 2026

Understanding the Human(s) in Human-Robot Collaboration: A Socio-Technical Inquiry into Robot-Assisted Surgical Work

HCC Seminar Series, Aalborg University, DK
March 2026

The Choreography of Care: An Ethnographic Study of Human-Robot Collaboration in Makoplasty Surgeries

QCR Seminar Series, QUT Centre for Robotics, AU
March 2026

Exploring Human-Robot Collaboration in Surgery: Initial Insights from Mako Product Specialists

ACM OzCHI 2025 Conference, Sydney, AU

December 2025

Removing the Drapes: Reflexive Insights on Research into Real-World Surgical Human-Robot Collaboration

ACM OzCHI 2025 Conference, Sydney, AU

December 2025

Opening the Theatre Doors: Observing Human-Robot Collaboration in Surgical Practice

HDR Seminar Series, School of Architecture and Built Environment, QUT

November 2025

Understanding and Designing for the Human(s) in Surgical Human-Robot Collaboration

HDR Symposium, Faculty of Engineering, QUT

October 2025

PUP116 - Human Factors & Ergonomics

Guest Lecture, Faculty of Health, QUT

September 2025

Read Them Sideways: Robotically-Assisted Surgery with Jasper Vermeulen

QUT Digital Media Research Centre, AU

June 2025

Human-Robot Collaboration in Surgical Robotics

Technical University Eindhoven, Human-Technology Interaction Group, NL

May 2025

Industry 5.0 in Surgery?

Erasmus University of Rotterdam, SEISMEC Group, NL

May 2025

Exploring Human Performance in Mako-Assisted Hip Replacement Surgeries

Chartered Institute of Ergonomics and Human Factors (CIEHF) Annual Conference, UK

April 2025

Human-Centred Robotics in Total Arthroplasty Surgeries

Queensland Orthopaedic Research Fund (QORF), Annual Meeting, AU

April 2025

Investigating human factors in Mako-assisted Total Knee Arthroplasty surgeries

ACM/IEEE HRI 2025 Conference, Melbourne, AU

March 2025

PhD Research Overview Presentation

Australian Computer-Human Interaction Conference 2024, Doctoral Consortium, AU
December 2024

Why is Human Factors important in Robotics?

Human Factors and Ergonomics Society of Australia (HFESA), Annual Conference, AU
November 2024

"Meet Your Mako: Exploring Human-Robot Collaboration in the Operating Room"

QUT 3MT Competition, Faculty of Engineering Final, AU
July 2024

Leadership & Service

Invited: Australian Computer-Human Interaction Conference 2026 Committee Member, Publications Co-Chair 2026 (Present)

Together with Dr Joel Harman, I am responsible for the publication pipeline of the OzCHI2026 conference.

Organiser and Presenter, Socio-Technical Grounded Theory in HRI: A Hands-on Tutorial for Coding and Memoing, IEEE RO-MAN 2026 (Present)

Building on the work of Prof Hoda, I will deliver a tutorial at IEEE RO-MAN 2026 on applying the Socio-Technical Grounded Theory for data analysis.

Elected: SIGCHI Committee Member: Industry Co-Chair 2025 – Present

Part of the Special Interest Group for Computer-Human Interaction (SIGCHI) Committee. My role focuses on connecting the Brisbane CHI community with relevant industry professionals, primarily Human Factors professionals and designers.

Nominated: Australian Cobotics Centre (ACC) PhD Representative 2025–Present

Represented ACC PhD candidates in centre-level meetings, supporting student engagement and feedback processes.

Elected: Higher Degree Research (HDR) Representative, Research Degrees Committee 2025–Present

Advocating for PhD student interests in university research policy discussions and programme development.

Organizer and Presenter, Practical Workshop on How to Use Atlas.ti for Qualitative Data Analysis, QUT Digital Media Research Centre Annual Summer School

2024–2025, 2025-2026

Together with PhD Researcher Mal Fituch from the Faculty of Law, I designed and delivered a workshop at the DMRC summer school, teaching participants how to use the Atlas.ti software for qualitative coding in their PhD data analysis.

Elected: Chartered Institute of Ergonomics and Human Factors (CIEHF) Annual Meeting (PhD) Student Volunteer

2025

Together with other student volunteers, we supported the organising committee by ensuring all the sessions ran smoothly.

Nominated: ACC Symposium Organising Committee Member

2025

Contributed to the planning and execution of the annual ACC symposium, including speaker coordination and event logistics.

Invited: Organiser and Presenter, Virtual, Augmented, and Mixed Reality for Human-Robot Interaction (VAM-HRI) Workshop, ACM/IEEE HRI 2025 Conference

2024–2025

Led the in-person session of the VAM-HRI workshop at the HRI 2025 Conference in Melbourne, coordinating with international researchers and facilitating discussions on immersive technologies in human-robot interaction.

Elected: QUT Centre for Robotics (QCR) PhD Representative

2024–2025

Acted as liaison between PhD researchers and QCR leadership, facilitating communication and organising PhD community events.

Elected: Australian Computer-Human Interaction Conference 2024 (PhD) Student Volunteer

2024

Together with other student volunteers, we supported the organising committee by ensuring all the sessions ran smoothly.

Elected: Faculty of Engineering Student Representative, Student Representative Council (SRC)

2023–2024

Elected representative advocating for engineering students' interests within the QUT Student Guild, participating in policy discussions and decision-making processes.

Volunteering

Invited: **FIRST Lego Robotics Competition Judge (Brisbane, Australia)**

2024 – Present

The children (6-12 y/o) present their robot project to the other judges and me. We give the children marks and provide some constructive feedback.

Languages

- Dutch (Native)
- English (Near Native)
- German (Conversational in context)
- Spanish (Basic Proficiency)
- Japanese (Elementary Proficiency)

Awards

- **Research Achievement Award 2026** (Awarded by QUT Centre for Robotics)
- **HDR Success Spotlight 2026** (Awarded by QUT Faculty of Engineering & Science)
- **DIS Best Paper Honourable Mention Award** (Awarded by ACM DIS2026)
- **CHI Best Paper Honourable Mention Award** (Awarded by ACM CHI2026)
- **Centre Supporter 2025** (Awarded by Australian Cobotics Centre)
- **Contribution to Public Debate 2025** (Awarded by Australian Cobotics Centre)
- **Recognition Award for Seminar Delivery 2025** (Awarded by QUT School of Architecture and Built Environment)
- **HDR Success Spotlight 2025** (Awarded by QUT Faculty of Engineering & Science)
- **Emerging Leader Award 2024** (Awarded by Australian Cobotics Centre)
- **Most Radical Research Funding Pitch Award 2024** (Awarded by Australian Computer Human-Interaction Society)
- **Industry Research Collaboration Award 2024** (Awarded by Australian Cobotics Centre)
- **EPIC Centre Citizen Award 2023** (Awarded by Australian Cobotics Centre)
- **Best Undergraduate Thesis Nominee** (Awarded by Erasmus University of Rotterdam)